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A fast serial computation approach for computing viewshed of a region by detecting key spots on the terrain

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ABSTRACT

Computing a viewshed for a space (a polygon), rather than a single location (a point), requires analysis of the maximum terrain extent observable from any location within the space. For a digital elevation model (DEM), it is the sum of viewshed analyses for all the pixels inside the confined space. As the raster spatial resolution increases, the process becomes computationally expensive. In this study, we propose a new serial computation approach, named Regional Viewshed by Aspect Peaks (RVAP), in which key pixels that define the cumulative visibility map on a digital elevation model (DEM) are found, making the process computationally affordable. We compare our proposed method next to 1) a 5-cell grid interval method 2) and an all-encompassing but slow complete approach where the viewshed is computed for all the pixels and then overlaid. Six regions-of-interests (ROIs) on two Shuttle radar (~555 × 555 km dimensions, 90-meter resolution) DEMs are selected and categorized by their terrain ruggedness index (TRI) (i.e. slightly rugged, mildly rugged, highly rugged). The proposed approach is much faster, while its coverage is identical to the all-encompassing approach. Moreover, as the TRI increases, the accuracy of the proposed algorithm is not diminished, while its execution time decreases.

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Viewshed; viewshed of a region; visibility algorithm; terrain aspect; GIS

Introduction

Viewshed of a location may be defined as the distribution of the area of surrounding region visible from a location and has plenty of applications, such as in geography, planning, archaeology, defence, telecommunication, and more (Statuto et al. 2016, 2017, 2019; Chamberlain 2020). 'Total viewshed', or 'inherent viewshed' offer a representation for the visual properties inherent in a landscape based on its topography and are produced by summation of viewsheds generated from all of the cells (taken to represent possible viewpoint locations) of a digital elevation model (DEM) (Travis et al. 1975; Brughmans et al. 2018; J. Wang et al. 2000a). Total viewshed expands the capabilities of simple viewshed

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analysis by offering us the view size of each location on a DEM. In this way, a score is calculated for each pixel; the higher the score, the larger the view the pixel could observe (Wu et al. 2020; Kuna et al. 2022). Path viewshed is another concept by which the total area that a moving entity can observe during its journey on a given path (line) is computed (Stucky 1998; Ruzickova et al. 2021). It is the summation of multiple viewsheds along the path at a given sequence of distance or time (Cimburova and Blumentrath 2022). As the sequence gets smaller, the number of locations for which viewshed is calculated increases, so does the computational complexity (Chamberlain and Meitner 2013).

Calculation of a visibility map (VM) even for a single observer is very expensive and has a complexity greater than $O(N)$, where N is the total number of cells in a DEM. This means, computing the complete VM at all the cells of the terrain has a complexity greater than $O(N^2)$ while the inherent complexity of the problem is up to $O(N^3)$ (Sanchez-Fernandez et al. 2021). Some studies propose solving this limitation by parallel computing of simple single observer viewshed algorithms on multiprocessor systems (Mineter et al. 2003; Llobera et al. 2010). Albeit, along with the benefits of consecutive advancements in processing power, powerful servers, and web services offering both data and various types of visibility algorithms, the calculation complexity problem is going to be solved (Clematis et al. 2003). But from 2005 onwards, there have also been various hardware and computational barriers in improving processor speed (National Research Council 2011: 151). These complications justify the need for introduction of new ways for speeding up the process.

The aim of this study is to introduce a fast and accurate approach to get the visibility of a region, instead of a single observer. It differs from the concept of total viewshed analysis. In total viewshed, as it is expressed earlier, the visibility map of each pixel is calculated and the outputs are totalized. So for the final output, pixels with higher values indicate that they are seen from many other pixels, while lower values indicate they are the least visible (Dungan et al. 2018). For this very reason, we've chosen 'viewshed of a region or an area, or a closed place' for it is a better term. 'Region's viewshed', or 'viewshed of a region' are alike and mean: 'the cumulative area that is visible from all cells within an enclosed region'. This is the least studied subject in the visibility literature. Owning to the fact that the simplest way to compute the viewshed of a region is calculating a series of viewsheds for all of the region's cells (pixels) and merge or overlay the outputs together. This will show the visible terrain, no matter where exactly the observer stands inside the observer region. But a small region might contain thousands or even millions or more cells on modern digital elevation models (DEMs) (Hao-Nguyen et al. 2018a; Hao et al. 2019). This would cost a lot in terms of processing power and also time (He et al. 2018). So calculating this heavy process is sometimes a challenge because of the intricacy of the computation and the current speed of computer processors, and this is progressively challenging over time as elevation data becomes increasingly fine in resolution (Tabik et al. 2013; Moll et al. 2020; Qarah 2020; Sanchez et al. 2020; Sobala et al. 2020).

This study makes possible a range of different advantages for various situations. For instance, to calculate the visibility of a whole city, village, or a site regarding its outer space, wherever the observers stand; to set up defence properties based on how much visibility a region has over a particular area; To plan for hiding an unpleasant scenery from a region or vice versa; and even to select the best overlooking regions in planetary exploration missions (Mahanti and Robinson 2014). The new approach recognizes visibility obstructions inside the region and identifies key locations to make the process a lot faster and less consuming.

Background

Terrain visibility, which is the same as viewshed analysis, is related to the determination of the visible terrain from a given point of view (POV) located at a certain elevation above the

surface. This issue has been widely studied for many years by applying various calculative procedures (Atallah 1983; Cabral et al. 1987; Fischer 1992; Floriani and Magillo 1994; Franklin 1994). That POV is called the observation point, and all the surrounding cells are called target-points (Ben-Shimol et al. 2007). According to Pan et al. (2020): 'there are two main methods of viewshed computation: the reference planes method and the line-of-sight (LOS) based viewshed computation method'. At present, it is common to use the LOS. Amongst these calculative procedures, R2 (by Franklin 1994), R3 (by Shapira 1990), Sweep-line (by van Kreveld 1996), and DDA (by Maĭorov 1964) algorithms are the most popular (Kaučič and Zalik 2002). There are other LOS variants as well such as XDraw, and High-Level Direction (HLD) that are faster regarding either how they interfere with the hardware or their procedure (Cauchi-Saunders and Lewis 2015; Hao-Nguyen et al. 2018b; Wu et al. 2020). But all these methods, basically, draw sightlines, from the observer toward the DEM boundary, where each sightline is a line-of-sight function that computes visibility for that direction. The difference between these algorithms is how much memory they use and the logic in which information gets processed and stored (Toma 2012).

Other algorithms use the concept of 'reference planes' instead of 'sightlines' for deriving the viewshed for a given viewpoint. The idea was first introduced by J. Wang et al. (2000b) in which a reference plane is defined as the plane passing through the viewpoint and a pair of adjacent cells lying on either the same row or column of an auxiliary grid. This algorithm is much faster than most algorithms that use sightlines.

Regardless of using neither sightlines nor reference planes, Wu et al. (2020) proposed a different algorithm built around a new idea. Proximity Distance Elevation Line Reference (PEDRL) advantage is that the spatial linear relationship of lines in geographic space can be transformed into the linear relationship of direction, elevation, and proximity, which lays the foundation for a fast and accurate viewshed analysis. Furthermore, a PDE spatial reference line algorithm (PDERL) is proposed based on the above coordinate system.

The explained LoS calculation has been the backbone of viewshed algorithms for years where most of them, calculate the viewshed from a single point-of-view, or a small number of POVs at best (Shen et al. 2008). During the past few years, researchers have also made these algorithms to run on graphics processing units (GPU) instead of central processing units (CPU) (Cauchi-Saunders and Lewis 2015; Green et al. 2012; Pan et al. 2020; Starek et al. 2020; Y. Wang and Dou 2020). This improvement speeds up the process times faster (Kang et al. 2018). This novelty lets analysts calculate the viewshed for multiple observers simultaneously (Stojanovic and Stojanovic 2013). Moreover, calculation of viewshed for all cells inside a region (or total viewshed) is now possible. But even now, too many raster cells could slow down the process to hours or days (Strnad 2011; Xiao-dong 2013).

In addition, there have been more visibility or intervisibility approaches developed recently. Namely, 'Voxel Viewshed' and 'the R.survey'. Voxel viewshed algorithm, as the name suggests, treats an elevation model as an array of three-dimensional discrete elements. Then the elements are categorized in such a way that non-observable elements are eliminated at the beginning. Therefore, the remaining elements are visible to the observer. In the voxel surface model, each of the raster cells in a digital terrain model is projected as a series of vertical columnar elements whose vertical and horizontal plains can be surveyed for partial visibility. It delivers 98% accuracy but on up to 1000× processing speed (Washtell et al. 2009; Fisher-Gewirtzman et al. 2013). The R.survey approach relies on three visibility metrics: 1. The distance to the target 2. Maximum view angle 3. How much it can be perceived. It also provides outputs not achievable by other similar packages, such as 3 D distance or view angle, and visible object's size (Sahraoui et al. 2018;

Bornaetxea and Marchesini 2022; Inglis et al. 2022). However, all these approaches are developed for single or double-observer scenarios.

By and large, fewer studies address the problem of a complete cumulative viewshed computation. And, they either try to lower the DEM resolution (Cervilla et al. 2017; Brughmans et al. 2018), lower the observer cell grid resolution (Dungan et al. 2018), or thicken the LoS to a band-of-sight (BoS) for a greater inclusion (Tabik et al. 2013, 2015). All are done to achieve a reasonable speed in executing the computation. However, these approaches have important limitations:

- For lowering either DEM or cell-grid resolution, there is a big chance of missing important peaks or bottoms that can obstruct or expand the view for many of the surrounding cells.
- By using the BoS technique, since the band rotates and sweeps sequentially, a vast amount of area gets recomputed every time; thus lowering the efficiency.

Viewshed of a region computation is a sequence made of many single-point (single-observer) viewshed algorithms which gives binary results (i.e. visible or not). This concept has not been fully studied yet. This technique is used to compute which areas are visible from a region instead of a POV. Meaning, wherever the observer stands inside a confined space, which parts (pixels/cells) outside of that space is invisible to his eyes. Its main limitation is for larger regions where the viewshed needs to be computed over and over again. For this reason, in every research that's been done over large regions, a type of either observer randomization or cell-skipping is used (Zamora 2010; O'driscoll 2017). But due to the skipping process, a lot of key spots, that are influential to the final visibility map, are randomly skipped (Kooistra 2019).

Here is a new method on how we could compute viewshed of a region using terrain aspect analysis (Horn 1981; Mitásová and Hofierka 1993; Dunn and Hickey 1998; Hickey 2000) to identify key locations and construct a serial computing procedure that is fast and also accurate.

Material and methods

Study area

The study region consists of a couple of standard blocks of 90 m spatial resolution, each with the extents of 5×5 degrees ($\sim$250000 Km^2), Shuttle Radar Topography Mission (SRTM) digital elevation model. The first one (SRTM 15-03 block) contains the northern parts of Montana State, United States of America, also southern parts of Alberta and Saskatchewan States of Canada. The second one (SRTM 13-03 block) contains the north-western parts of Montana State, northern parts of Oregon, Idaho, United States of America, also southern parts of British Columbia, Canada; since they are typically found together as a standard 90 m digital elevation model (DEM) file which is easily accessible.[1] The study region covers diverse landscapes, from extremely rugged Kootenay ranges, steep mountains, gentle hills, to plain and smooth lands. Figure 1 shows the entire coverage of our study region. The observer for 'viewshed of a region' is not a cell, but all the cells which are located inside an area (a polygon). Thus, it is better to be called the observation region. Therefore, all the opposite or outside surrounding cells could be called target areas accordingly.

We select six regions of interest (ROIs) as our observation regions based on the terrain ruggedness index (TRI). The terrain ruggedness index shows the sharpness of elevation

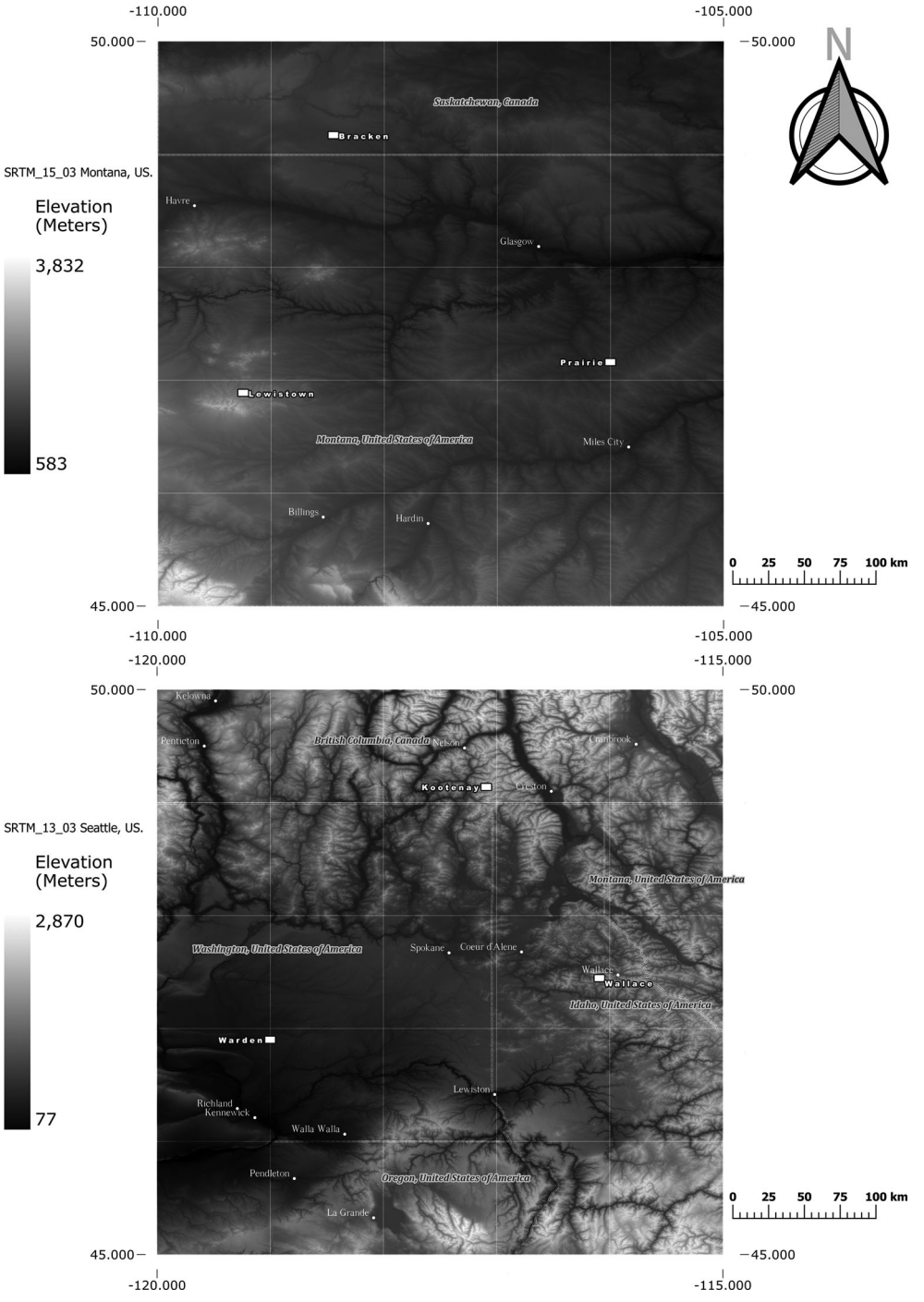


Figure 1. Study regions of interest. Six regions of interest (ROIs) are specified on two 5 × 5 blocks of standard Shuttle Radar Topography Mission (SRTM) digital elevation models, SRTM 13-03 and SRTM-15-03. the six designated ROIs are named Bracken, Prairie, and Lewistown for SRTM 15-03 and Warden, Wallace, and Kootenay for SRTM 13-03.

vicissitudes for a specific portion of space. Therefore, it gives information about the steepness or plainness of a region. It is seriously important for a viewshed algorithm to analyze any type of landscape quickly and accurately (Riley et al. 1999). Higher TRI equals to higher terrain complexity and this is an influencing part in assessing the computational complexity. All of the ROIs have the same width and length: 0.058 degrees of horizontal width ($x \approx 6500$ m) and 0.062 degrees of vertical length ($y \approx 7000$ m). Doing so results in ROIs with various number of pixels caused by difference in regional elevation. We name six three ROIs according to the name of their closest habitation. Bracken is the slightly rugged ROI, while Lewistown is a highly-rugged. And the Prairie ROI is something in between. These are for the first DEM (SRTM 15-03). For the second DEM (SRTM 13-03) Wallace, Warden, and Kootenay as slightly rugged, somehow rugged, and extremely rugged in order. Figure 2 shows the TRI map of the region and the selected ROIs are drawn on it.

Proposed algorithm

'Viewshed of a region' could be intuitively imagined as the overlay of all the viewshed outputs for each and every cell inside the region of interest. Although the computation intensity for such a task, especially where the observer region contains a large number of cells (e.g. more than a thousand cells), is tremendous and could easily be hours to days of constant computation. A simplified representation is shown in Figure 3.

Because topographies are made of changes in slope direction, rationally conceptualizing, it is the main driving force behind any change in the viewshed output (Varien 1999: 141). The proposed approach is a serial computing algorithm and is constituted of J. Wang et al. (2000a, 2020b) Reference Planes viewsheds based on the ROI's slope direction (i.e. aspect), also the local maximum elevations inside each one of the aspect patches.[2] The reason behind developing the algorithm using Reference Planes viewshed, and not faster methods such as the GPU-implementation, is its availability in most software development kits (SDK) and computer programming libraries. It facilitates greater interoperability for future research and utilization.

Each aspect map is a classification of slope directions for a terrain. The smaller constituents of an aspect map are groups made of identical pixel values, referring to identical slope direction of the pixels, and are called 'aspect patches'. Therefore, an aspect map is composed of many aspect patches. Figures 4 and 5 show the procedure based on which the proposed algorithm operates. We named this algorithm Regional viewshed by Aspect Peaks (RVAP)[3] for the purpose of reading convenience.

It is contemplated that it is possible to calculate the viewshed of a region, by discovering the key points (cells) inside the ROI and calculating viewshed analysis for these few key points instead of all the cells (pixels) of the ROI. And discovering these key points is possible if we, first, calculate an aspect map for the terrain. Second, find the maximum elevation cell inside each of the resulted aspect patches. Third, as numerous tiny spots are created by aspect computation, the elimination of these negligible spots is essential for the purpose of faster computation. But it is only permitted if no pivotal elevation is located inside them. So non-key points and their relatively small patches are eliminated as their viewshed is subjacent to those of the neighbouring patches.

Elimination of the negligible aspect patches is very similar to constant median image filters that are used in image editor software packages. Where the undesirable patch is not removed (i.e. replaced by no-value or zero), but it is torn into parts. Then, each part will be equally replaced by the value of its neighbouring patch. After these steps, tall bumps

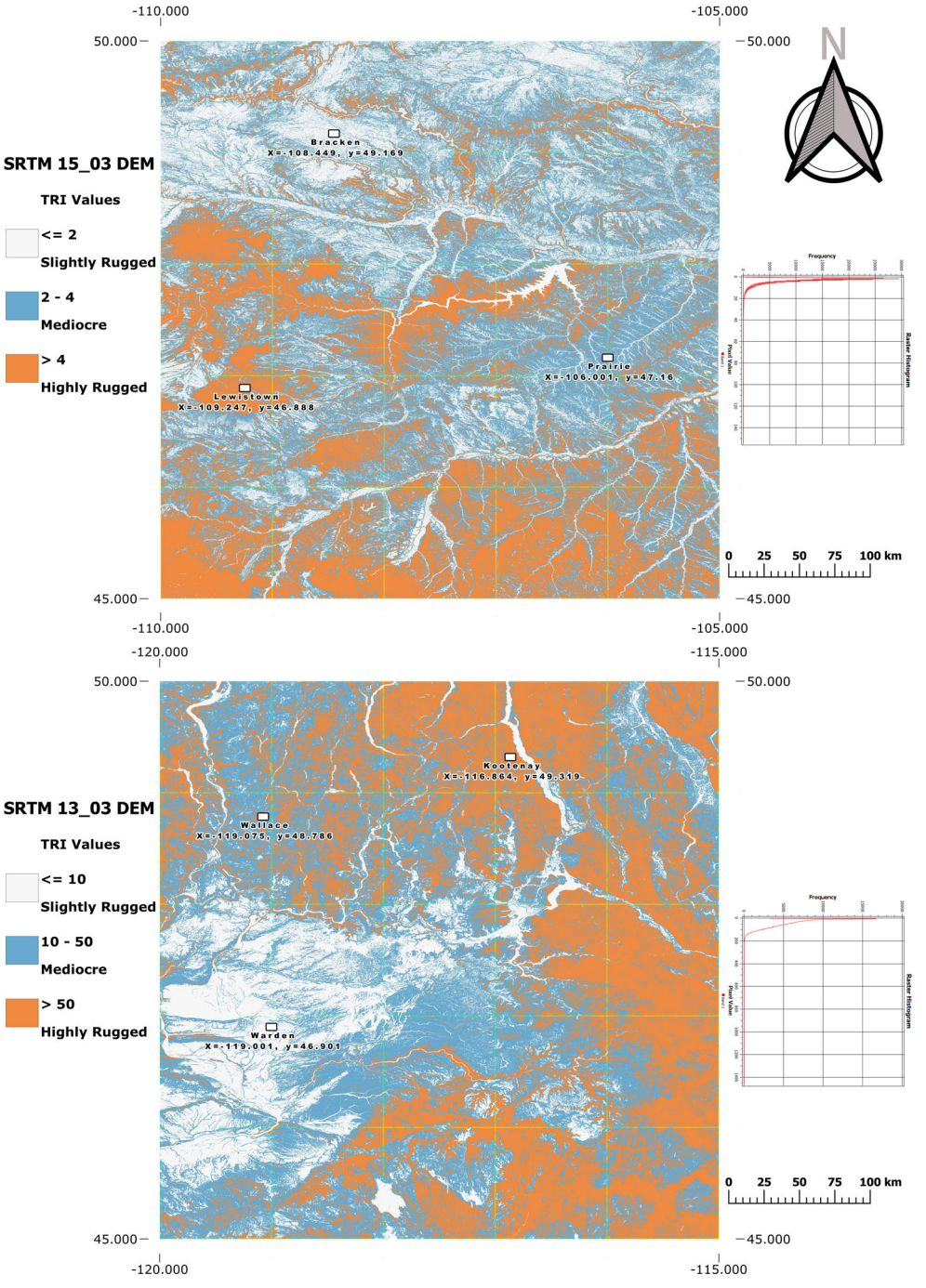


Figure 2. Terrain ruggedness index map and the designated regions of interest (ROIs). Ruggedness values are generally higher for SRTM 13-03 digital elevation model indicating its landforms are among the most rugged places in the world. On the right side of each figure, raster histograms are drawn. Lower pixel values (<20 for SRTM 15-03 and <200 for SRTM 13-03) are the most frequent; suggesting that major parts of the elevation models are smoother surfaces in general.

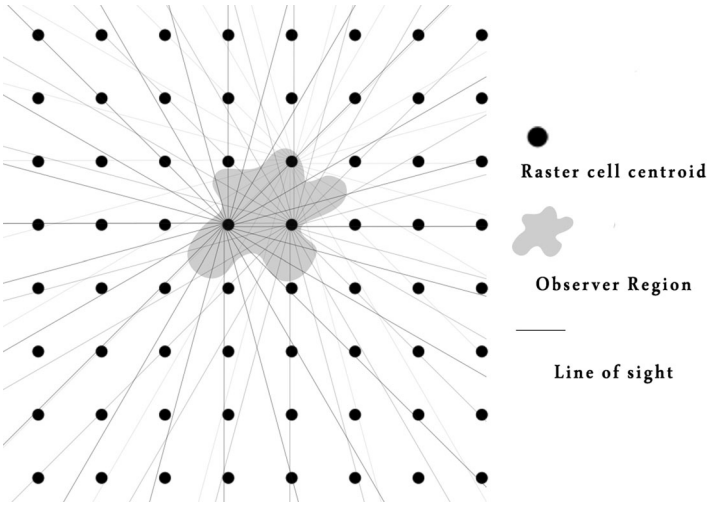


Figure 3. A simplified illustration of common Viewshed of a region based on a rational conceptualization.

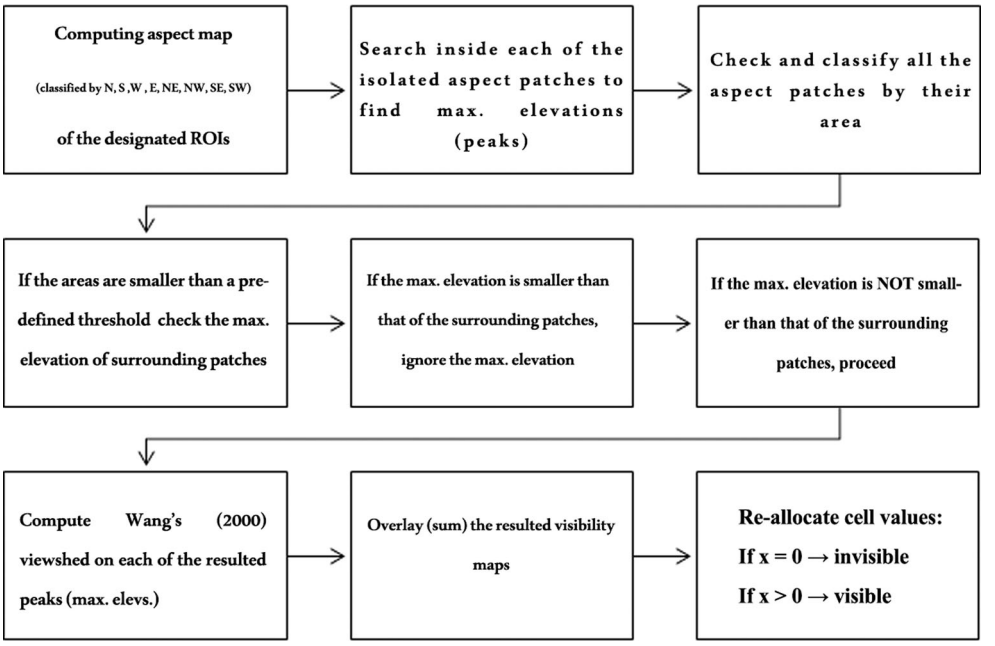


Figure 4. A diagram that shows the procedure for the proposed algorithm.

and also large areas of the same slope direction will remain as their peak tops could augment the final viewshed, that could not be done from any other observer region. And this helps to maximize the final visibility status. Thus, the main contribution of our method is this elimination phase that makes the results more constant. This is illustrated in step 3 of Figure 4. We set the elimination threshold to less than 24,300 square meters or the combined area of just three 90 m DEM pixels. This means any aspect patch with an area smaller than 24,300 m^2 having no maximum elevation higher than that of the surrounding patches is eliminated. Wet set this threshold because of the scale and precision of the input data.

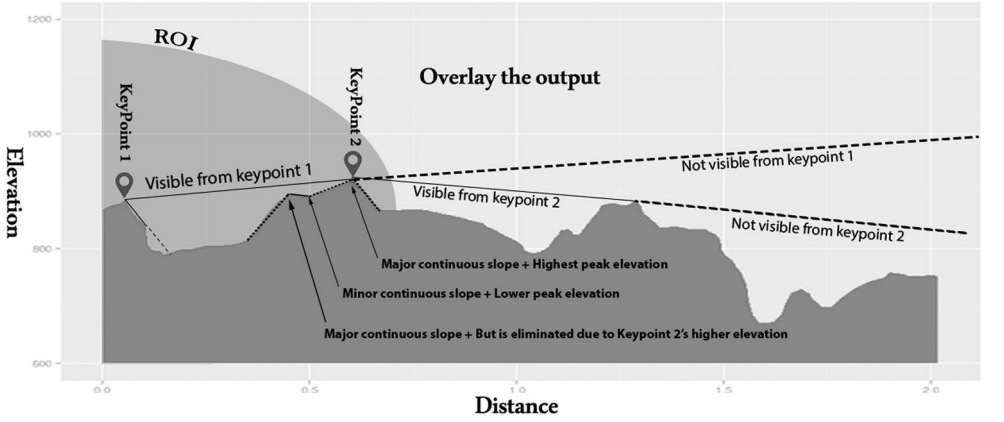


Figure 5. Key points selection inside a region of interest (ROI).

The proposed approach in terms of mathematical notation is as:

$$a_{i,j} \in R^{m\times n} \because A = \begin{bmatrix} a_{0,0} & \cdots & a_{0,m} \\ \vdots & \cdots & \vdots \\ a_{0,n} & \cdots & a_{m,n} \end{bmatrix}$$

$$a_{i,j} \in \begin{cases} \gamma_{i,j} \in \{y_0 \cdots y_n\} \\ e_{i,j} \in \{e_0 \cdots e_n\} \end{cases}$$

Where a is a vector of the matrix A and m, n are the 'column, row signifiers' or 'matrix dimension decorations'. The succession for a large aspect patch is an *if* statement:

$$(x_{i,j}) = \begin{cases} pass, \\ viewshed_{(e.\max_{\gamma_{i,j}})}, \end{cases} area_{\gamma_{i,j}} > 3 * cellsize$$

The elimination of small and non-influencing patches and their relative maximum elevation:

$$(x_{i,j}) = \begin{cases} pass, & area_{\gamma_{i,j}} < 3 * cellsize \\ eliminate(\gamma_{i,j}) & e.\max_{\gamma_{i,j}} < e.\max_{\gamma_{i\pm1,j\pm1}} \end{cases}$$

The succession for a small but pivotal aspect patch:

$$(x_{i,j}) = \begin{cases} pass, & area_{\gamma_{i,j}} < 3{*}cellsize \\ viewshed(e.\max_{\gamma_{i,j}}) & e.\max_{\gamma_{i,j}} > e.\max_{\gamma_{i\pm1,j\pm1}} \end{cases}$$

Where R is a hypothetical image array with the same dimensions as our DEM or the consequential aspect map. $\gamma_{i,j}$ is an element (patch) on the aspect image where i, j is its x, y coordination. $e_{i,j}$ is an element (pixel) on the elevation model using the same coordination but $e.\max_{\gamma_{i,j}}$ is the maximum elevation inside $\gamma_{i,j}$ patch.

Here we encapsulate the formula as a simplified and readable pseudocode block:

```
# → the comment sign
# 1- This part will reclassify the aspect patches to give them a unique ID number
  search every row in Aspect rows:
    search every column in Aspect columns:
```

reclassify the aspect into 8 values of N, NE, E, SE, S, SW, W, NW

Add elevation values to each pixel

So a pixel could have a value like NE,8000 –

NE for slope direction and 8000 for elevation

2- This part will find the maximum elevation in each raster patch

find the maximum elevation in each aspect patch (not pixel) e.g. NE, xxxx

3- Check if elevation intensity and aspect coninuity is validated

if patchsize $<$ threshold area

and if maximum elevation (peak) $<$ (max elevation of surrounding patches + the observer height):

delete peak and eliminate the patch

Validation

Dungan et al. (2018) showed in their research the idea of placing 'grid-intervals'. This means that to increase a cumulative viewshed analysis's (CVA) speed, we could place a 5-cell interval, instead of computing each and every cell, while still getting somehow accurate results. This idea makes the computation (n) times faster. Where (n) is the grid-interval, so, in the case of this example, is (n^2) times faster.

Here, we compare 1) the 5-cell-interval method, 2) the proposed aspect-based method, and lastly 3) a simple serial computing approach that computes viewshed for all the ROIs' pixels and overlays the results at the final stage. And it is self-evident that the latter approach must be the slowest but the most accurate. The inclusion of the 5-cell method is because, like the proposed RVAP algorithm, it is an attempt for increasing the speed by reducing the number of viewsheds that will be added to the final composite. We apply these three different approaches to the six designated ROIs and compare the results. Assessing the accuracy and the speed of the proposed approach by juxtaposing its results to those of both the complete serial procedure (or the complete cumulative approach), as the most accurate, and the grid-interval (as a benchmark algorithm) could be a solid validation stratagem. For all of these approaches, we use: 1.8 meter as the *observer elevation* (viewing elevation above the ground) during both the elimination and the main computation phases, 50 km as the maximum sight distance for the viewing cells/region, and 0.85714 as the *curvature coefficient*.

Results and discussion

The results, as shown in Figure 6, indicate the final output of the proposed algorithm (RVAP) versus the 5-cell grid-interval method and the complete cumulative procedure. Table 1 shows the comparison data between the three approaches on different ROIs. The benchmark (speed) for the complete cumulative approach ranges from $\approx$ 6 hours to $\approx$ 6 hours 23 minutes, varied because of the POV counts; as all the pixels were calculated. The benchmark for the 5-cell grid-interval method is ranging from $\sim$14.52 minutes for the slightly rugged ROI to $\sim$16.00 minutes for the mid-rugged ROI, then dropped to $\sim$14.54 minutes for the highly rugged. Again, the reason is that Prairie ROI was constituted of more pixels. Therefore, for the 5-cell grid interval method 1/25 of all the pixels were calculated. While the benchmark for RVAP ranges

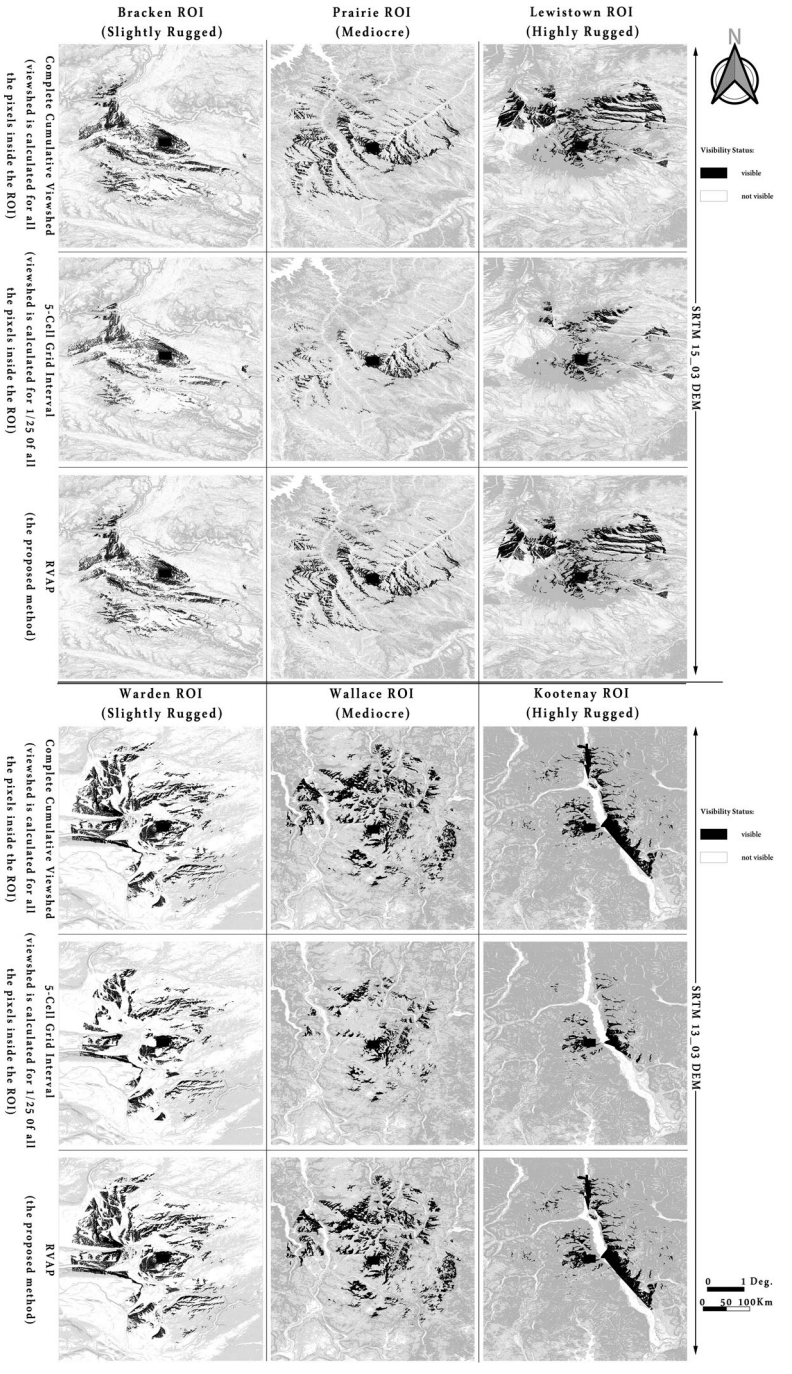


Figure 6. Final comparison for viewshed of a region using sequences of viewshed analyses done for each of the ROIs using complete cumulative approach, 5-cell grid-interval approach, and the proposed approach (RVAP).
The ROIs are Bracken, Prairie, Lewistown that are located inside SRTM 15-03 digital elevation model extents, and Warden, Wallace, and Kootenay that are located inside SRTM 13-03 digital elevation model extents.

from ≈ 1.29 minutes for the slightly rugged ROI to ≈ 1.43 minutes for the highly rugged ROI. It seems the scene size does not affect the RVAP's benchmarks, but the terrain ruggedness index does. Both algorithms only use the CPU for their computation.

Table 1. Comparison data between the proposed approach (RVAP), the 5-cell-grid-interval, and the complete cumulative approach on the desired ROIs.

Method \ ROI	Bracken (slightly rugged)			Prairie (mid-rugged)			Lewistown (highly-rugged)		
	Time	Accuracy compared to the complete cumulative	POV Count	Time	Accuracy compared to the complete cumulative	POV Count	Time	Accuracy compared to the complete cumulative	POV Count
Complete viewshed	~359 min	–	7560	~399 min	–	7668	~376 min	–	7597
5-cell grid Interval	~14.52 min	75.1%	302	~16.00 min	53.7%	307	~14.54 min	32.3%	303
RVAP	1.29 min	100%	33	1.35 min	100%	49	1.43 min	100%	63
Method \ ROI	Warden (slightly rugged)			Wallace (mid-rugged)			Kootenay (highly-rugged)		
Complete viewshed	~339 min	–	7522	~321 min	–	7514	~396 min	–	7634
5-cell grid Interval	~14 min	42%	301	~14 min	21%	300	~13.48 min	18%	296
RVAP	1.46 min	100%	66	1.11 min	100%	25	1.25 min	100%	30

All three approaches are implemented using CPP, Python computer programming languages.

The applied computer hardware for this research is AMD FX 8370 and 8GB DDR4 Gigabyte memory (RAM).

Calculating viewshed analysis for all the cells inside an ROI would exponentially increase the benchmarks compared to the 5-cell grid-interval the proposed approach. Because if you consider all five out of five cells, both horizontally and vertically, according to $A = \rho^2$, processing speed is decreased. It is worth mentioning that for higher resolution elevation models and larger study ROIs the calculation speed might be too slow and might take long to days or even months of constant computations. The complete cumulative approach acquires the highest achievable accuracy, the grid interval approach is faster, but it is sacrificing accuracy in favour of the speed.

By attending to Figure 6, both the complete cumulative approach and RVAP produce 33.8% more coverage for the Bracken ROI (slightly rugged), 86% more coverage for the Prairie ROI (mid-rugged), and 201.3% more coverage for the Lewistown ROI (highly-rugged) compared to the interval approach. Warden, Wallace, and Kootenay ROIs, that had generally higher TRI values or were more rugged in nature show the true advantages of RVAP. The coverage percentage values are calculated by: $x = \frac{(V_2 - V_1)}{V_1} \times 100$ where V_2 is the number of cells created by RVAP (or the complete cumulative approach) and V_1 is the number of cells computed by the 5-cell grid-interval algorithm. Comparing the proposed RVAP to the complete cumulative approach reveals the 100% accuracy of RVAP. Although the accuracy is calculated by dividing the coverage percentage between either RVAP or the 5-cell-interval and the complete cumulative approach.

Compared to an approach where a viewshed for all the ROIs pixels is calculated and overlayed to the rest, AVAP computes the viewshed of a region times faster but produces exactly the same results. The 5-cell grid interval approach falsely encompasses less area, or renders fewer pixels as visible, on the target region. According to Dungan et al. (2018), the 5-cell grid-interval should give coarser but somehow accurate results in computing cumulative viewshed analysis. But, when the cells are skipped randomly, a lot of the key points regarding their position and elevation get denied. This greatly sacrifices the accuracy and the coverage for getting a faster processing speed.

Where the terrain is rugged, it has more variation in elevation in the course of distance. And it produces more key points. Therefore, if we use a grid-interval approach we might lose many of these points. But if these key points are recognized, as done using the proposed method, and perform a viewshed serial computation for them, the observing points which could see hidden parts of the target region are all included in the process. Therefore, we would have much more accurate results.

The comparison between the complete cumulative, RVAP and the 5-cell grid-interval approaches (Figure 6) indicates that the results are compatible. RVAP covers all the area that is observable by the complete cumulative method, but the interval method fails to succeed at it. The reason is that the 5-cell grid-interval method skips many key points from which larger parts of the target region could be observed. For none of the ROIs, RVAP and the complete approach showed any difference in their results. The reason is that RVAP finds all the existing key points that when put together define the maximum visibility of the observer region. The visibility is always the same as that of the complete procedure.

Conclusion

This study illustrates the utility of a novel 'viewshed of a region' approach and demonstrates the degree to which the development of a viewshed algorithm, which employs aspect to find key points for each patch and direction, make this computationally

intensive technique practical. It has answered the question of 'how could we calculate the visibility of a confined space over its outer regions, instead of a sole observer, not by summing the visibility of each cell inside that space'. The essential method applied here (RVAP) can be replicated with other software packages, so that the application could reduce the analytical time, regardless of the platform used.

The applications of this research are vast. Think of finding an unobservable location in which waste materials are disposed, regarding the cities nearby. We don't know where the observer is located inside the cities' confines. Or when an army wants to transport their needs unseen near an enemy site. Also, for archaeology or even planetary exploration missions, where visual settings of a region or multiple regions are of concern. RVAP will allow us to do these analyses much more efficiently and by extension allow the analysis to be undertaken at very large scales that would have been unfeasibly time consuming previously.

Results indicate that by increasing the terrain ruggedness RVAP's accuracy is not diminished. A DEM with higher resolution, typically, has higher TRI values. Because the change in elevation by moving towards a direction is increased because of the higher resolution (more cells with regards to the distance). So, it is anticipated, that RVAP operates neatly on finer resolutions. But it is best to experiment on this in future research.

Notes

1. **Product:** SRTM 90m DEM Version 4
   SRTM 15-03:|**SRTM 13-03:**
   Latitude Min: 110 S **Max:** 105 S|**Latitude Min:** 120 S **Max:** 115 S
   Longitude Min: 45 E **Max:** 50 E|**Longitude Min:** 45 E **Max:** 50 E
   Center Point Lat: 107.5 S **Long:** 47.5 E|**Center Point Lat:** 117.5 S **Long:** 47.5 E
   Data could be freely downloaded at https://srtm.csi.cgiar.org/wp-content/uploads/files/srtm_5x5/TIFF/srtm_15_03.zip (Jarvis et al. 2008).
2. Wang et al. (2000a) algorithm could be replaced with any other viewshed algorithms available.
3. The algorithm could be accessed at http://doi.org/10.17632/7jwd5nm45n.1

Author contributions

All authors read and approved the final manuscript.

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Conflict of interest statement

The authors have no competing interests to declare that are relevant to the content of this article. And all authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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